

Question	Answer	Mark	Mark scheme	Additional guidance
20	3.75	P1	works to find vol of frustum eg $\frac{1}{3}\pi(3.6)^2 \times 6.4 - \frac{1}{3}\pi(1.8)^2 \times 3.2$ or $86.858.. - 10.857... (=24.192\pi \text{ or } 76.00..)$	781.7... by use of diameter does not get the mark [vol] is their volume which could be ft using the radius, using the diameter, or could be another value as long as it is stated as being the volume, or clearly intended from working. All figures must come from correct method shown.
		P1	works to find vol of hemisphere eg $\frac{1}{2} \times \frac{4}{3} \pi \times 3.6^3 (=31.104\pi \text{ or } 97.7....)$	
		P1	mass of frustum as [vol] $\times$ density eg “76.00” $\times$ 2.4 (=182.4.. or mass of hemisphere as [vol] $\times$ density eg “97.7....” $\times$ 4.8 (=469.037...)	
		P1	mean density as total mass $\div$ total volume eg (“182.4..” + “469.037”) $\div$ (“76...” + “97.7..”) or “651.4..”. $\div$ “173.7....”	
		A1	answer in the range 3.7 to 3.8	